

Spatial Fidelity of Grad-CAM in Chest Radiography

Quantitative Evaluation against Radiologist Annotations on VinDr-CXR

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Abstract

Saliency-map explanation methods, including Grad-CAM, are massively used in medical imaging to interpret the decisions of neural networks, but their spatial fidelity is rarely validated quantitatively against clinical ground truth. We evaluate here the ability of three CAM variants (Grad-CAM, Grad-CAM++, XGradCAM), applied at two depths of a ResNet-50 trained in JAX/Flax on 12,000 VinDr-CXR chest radiographs, to localize 14 pathologies annotated with bounding boxes by radiologists. The multi-label classifier achieves a macro area under the curve (AUC) of 0.920, demonstrating strong discrimination. By contrast, localization depends strongly on the pathology: the pointing game reaches 0.47 for cardiomegaly (large central lesion) but drops to 0.00–0.09 for nodules, pneumothorax and pleural thickening (focal lesions). Weight randomization tests confirm that cardiomegaly localization reflects learned features ($p = 2.86e-21$), whereas, for Grad-CAM, ILD (interstitial lung disease) localization is an architectural artifact ($p = 0.96$, not significant); this finding remains to be extended to Grad-CAM++, a configuration not controlled here. We recommend layer4 + Grad-CAM++ as the default configuration, the best of the six configurations evaluated without any being satisfactory for focal lesions, and document the clinical limits of these maps for micro-lesions.

1 Introduction

Interpreting deep learning models in medical imaging is a critical issue for clinical adoption. Class activation mapping (CAM) methods, and Grad-CAM [1] in particular, have become the de facto tool for visualizing the regions of a radiograph that influence a prediction. Yet these methods are frequently deployed without quantitative evidence that the highlighted regions correspond to the anatomical or pathological structures clinicians consider diagnostic. A recent benchmark has indeed shown that the seven saliency methods studied, Grad-CAM included, remain significantly below a human expert for the localization of thoracic pathologies [2]. The VinDr-CXR dataset [3] offers a rare opportunity to answer this question. It provides, in addition to classification labels, bounding boxes annotated by experienced radiologists for every detected pathology. These annotations constitute an independent spatial ground truth, never used during training, enabling an objective evaluation of the spatial fidelity of saliency maps.

The contributions of this work are as follows. We train a multi-label ResNet-50 classifier in JAX/Flax on VinDr-CXR and report its discrimination performance through per-class AUC. We systematically evaluate three CAM variants (Grad-CAM, Grad-CAM++, XGradCAM) at two depths (layer3, layer4) using four complementary localization metrics, with paired statistical tests (same images and pathologies) and false discovery rate (FDR, Benjamini–Hochberg) correction. We validate the robustness of the results through three controls: an untrained model, weight randomization, and random maps. Finally, we analyze the *No finding* class with diffusivity metrics and decompose all results by pathology, with sample sizes and confidence intervals.

Key Results

- Macro AUC of **0.920**: the classifier correctly discriminates the 15 classes.
- Max pointing game of **0.474** for cardiomegaly (layer4 + Grad-CAM++), but **0.000–0.087** for focal lesions (nodule, pneumothorax, calcification).
- Cardiomegaly localization survives weight randomization ($p = 2.86e-21$), ILD localization does not ($p = 0.96$), suggesting an architectural artifact for the latter.

2 Methods

2.1 Dataset and preprocessing

We use the VinDr-CXR dataset [3] (VinBigData competition, Kaggle), comprising 15,000 posteroanterior chest radiographs in DICOM format. Each image is annotated by a panel of radiologists for 14 pathologies with bounding boxes, plus a *No finding* class. Annotations are multi-label: the same image may present several pathologies simultaneously.

The split is deterministic (seed=42) into three disjoint sets: 12,000 training images (80%), 1,500 validation (10%) and 1,500 test (10%), the latter never used for training or model selection. The DICOMs are preprocessed once and cached in .npy format: rescale slope/intercept, windowing (or 1–99 percentiles), MONOCHROME1 inversion if needed, resizing to 224×224 . During training, ImageNet normalization and random horizontal flip are applied.

2.2 Model

The classifier is a ResNet-50 [4] in JAX/Flax NNX, trained from scratch (no ImageNet pretraining). The architecture explicitly returns the layer3 ($14 \times 14 \times 256$) and layer4 ($7 \times 7 \times 512$) activations in addition to the logits, allowing direct computation of CAM maps without hooks. The head is a dense layer with 15 outputs and independent per-class sigmoid (multi-label), optimized with averaged binary cross-entropy. The optimizer is AdamW [5] ($\text{lr}=10^{-3}$, weight decay= 10^{-4}), batch size 64, 30 epochs; the retained checkpoint corresponds to the best validation loss.

2.3 CAM methods

Three variants are evaluated. **Grad-CAM** [1] weights activations by the spatial average of the class-score gradients. **Grad-CAM++** [6] introduces a higher-order weighting (α) correcting the bias in the presence of multiple occurrences of the same class. **XGradCAM** normalizes by the sum of activations. Each map passes through ReLU, is normalized within $[0, 1]$, then upsampled by bilinear interpolation to the input resolution (224×224).

2.4 Localization metrics

For each positive class of a test image, the CAM map is compared to the corresponding bounding box (normalized by the original DICOM dimensions). Four metrics are computed: the **pointing game** (1 if the map's argmax falls inside the box), the **energy in box** (fraction of total activation contained in the box), the **IoU** (intersection over union between the map binarized at 0.5 and the box), and the **normalized distance** from the maximum to the box center, relative to its diagonal. When several radiologists annotate the same class, the best score is retained: the map is then considered correct if it coincides with at least one of the annotated regions, a convention that biases scores upward and constitutes a limitation. Since CAM maps are not calibrated segmentations, IoU binarized at a 0.5 threshold remains an imperfect metric; the continuous metrics (energy in box, distance) are authoritative for fine-grained comparisons. For the *No finding* class, without bounding boxes, we measure **diffusivity** through spatial entropy and concentration in the top 1% of pixels.

2.5 Statistical controls

Three controls verify that the metrics measure learned behavior and not an artifact: (i) an **untrained** model (random initialization), (ii) a model whose learned weights are **randomized** by permutation (destroys the features but preserves the distribution), and (iii) uniform **random CAM maps**. Control comparisons use the one-sided Mann-Whitney U test (real > null), reported globally and per pathology. Layer and method comparisons, for their part, use **paired Wilcoxon** tests (each finding is compared to itself), with **Benjamini–Hochberg** (FDR) correction per test family. The 95% confidence intervals are obtained by bootstrap with 2,000 replicates resampling images (not findings), in order to preserve the correlation between pathologies of the same image.

3 Results

3.1 Classification performance

The model reaches a macro AUC of 0.920 (mean of the 15 classes; 0.916 over the 14 pathologies alone) on the test set. The best discriminations are *No finding* (0.978), Aortic enlargement (0.972) and Cardiomegaly (0.966). The weakest remain correct: Infiltration (0.873), Calcification (0.878) and Nodule/Mass (0.880). Table 1 gives the AUC of each class. This performance level validates that the network has learned discriminative representations for the 14 pathologies, a necessary condition for the localization analysis to be interpretable. Figure 1 shows the training curves: the training loss decreases regularly over the 30 epochs, while the validation loss stabilizes around 0.15 after about ten epochs and remains noisy without marked divergence; the retained checkpoint corresponds to the validation-loss minimum.

Table 1: Per-class AUC on the test set. Macro over the 15 classes: 0.920 (0.916 over the 14 pathologies).

Class	AUC	Class	AUC	Class	AUC
Aortic enlargement	0.972	ILD	0.886	Pleural effusion	0.943
Atelectasis	0.948	Infiltration	0.873	Pleural thickening	0.931
Calcification	0.878	Lung Opacity	0.915	Pneumothorax	0.894
Cardiomegaly	0.966	Nodule/Mass	0.880	Pulmonary fibrosis	0.910
Consolidation	0.916	Other lesion	0.910	No finding	0.978

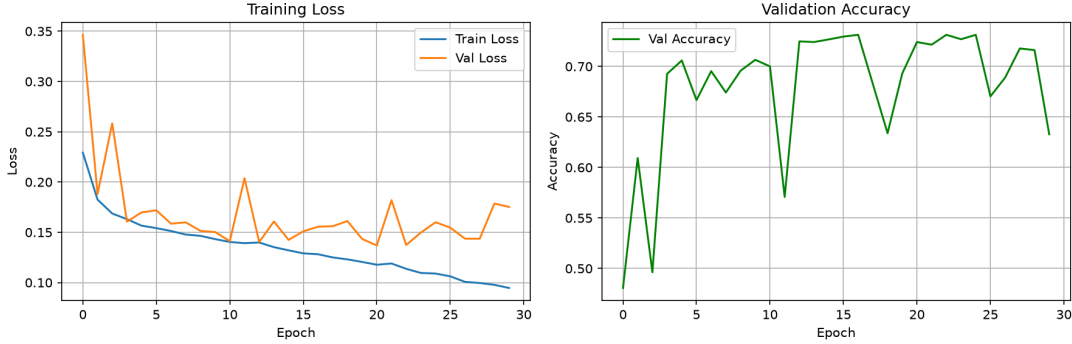


Figure 1: Training (30 epochs, seed 42). Left: training and validation loss (multi-label binary cross-entropy). Right: validation accuracy (threshold 0.5), noisy but stable around 0.70 after convergence.

3.2 Layer $\times$ method hierarchy

Table 2 summarizes the metrics of the two layers (mean $\pm$ standard deviation, $n = 442$ findings from 500 images). At layer4, Grad-CAM++ dominates: pointing game of 0.161 (95% CI 0.143–0.232) versus 0.109 for Grad-CAM, IoU of 0.091 versus 0.059; the differences are significant under the paired Wilcoxon test with FDR correction ($q < 0.002$). At layer3, however, Grad-CAM++ is the worst of the three methods (pointing game 0.007), which shows that its advantage depends on the chosen layer.

Table 2: Localization metrics by layer and method (mean $\pm$ standard deviation, $n = 442$ findings). Normalized distance: lower is better.

	layer3			layer4		
	Grad-CAM	Grad++	XGrad	Grad-CAM	Grad++	XGrad
Pointing game $\uparrow$	0.081 $\pm$ 0.274	0.007 $\pm$ 0.082	0.070 $\pm$ 0.256	0.109 $\pm$ 0.311	0.161$\pm$0.368	0.109 $\pm$ 0.311
Energy in box $\uparrow$	0.050 $\pm$ 0.072	0.033 $\pm$ 0.040	0.045 $\pm$ 0.060	0.052 $\pm$ 0.085	0.054$\pm$0.070	0.052 $\pm$ 0.085
IoU $\uparrow$	0.041 $\pm$ 0.073	0.029 $\pm$ 0.038	0.040 $\pm$ 0.077	0.059 $\pm$ 0.102	0.091$\pm$0.115	0.059 $\pm$ 0.102
Distance $\downarrow$	2.91 $\pm$ 3.50	3.68 $\pm$ 4.57	2.86 $\pm$ 3.70	2.74 $\pm$ 4.30	2.23$\pm$3.59	2.74 $\pm$ 4.30

XGradCAM and Grad-CAM coincide exactly at layer4 (maximum deviation $\leq 3 \times 10^{-6}$). This identity is not an implementation artifact: at layer4, the classification head is a global average pooling followed by a linear layer, which makes the score gradient **constant in space**; the XGradCAM weighting, which weights gradients by activations, then reduces exactly to the spatial average of the gradients used by Grad-CAM. The two methods diverge at layer3 (pointing game 0.081 versus 0.070), where the tail of the network is non-linear.

Figure 2 restates the same result graphically for the four metrics: each time, layer4 dominates layer3 for Grad-CAM++ and XGradCAM, and Grad-CAM++ at layer4 forms the maximum across the three increasing metrics (pointing game, energy in box, IoU) as well as on the distance (decreasing metric). Only Grad-CAM remains relatively stable between the two layers, consistent with the paired test of the following section.

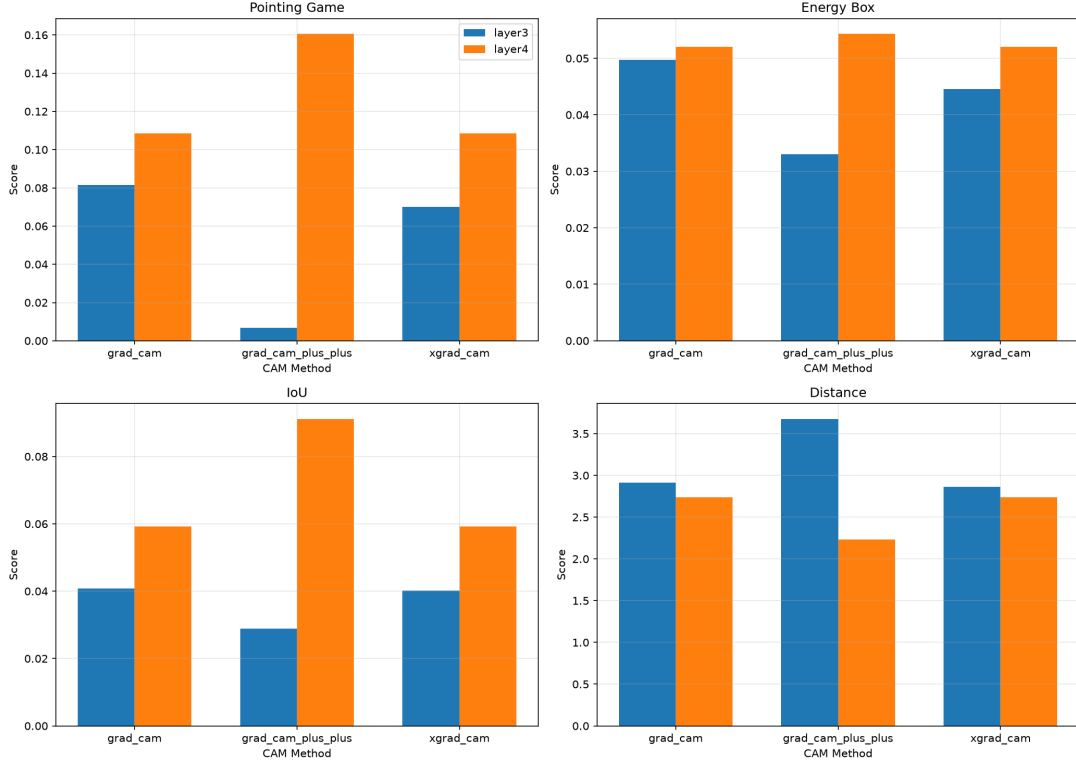


Figure 2: Means of the four localization metrics by method (x -axis) and by layer (color). Higher is better, except for distance (lower is better). Grad-CAM++ at layer4 dominates on all metrics; Grad-CAM++ at layer3 collapses (pointing game 0.007), showing that its advantage is conditioned on the layer.

The paired comparison of layer4 against layer3 (image $\times$ pathology pairs, 12 tests, Benjamini–Hochberg correction) confirms the superiority of layer4 for Grad-CAM++ (better on the four metrics, $q \leq 9 \times 10^{-46}$) and, more nuancedly, for Grad-CAM: IoU ($q = 1.2 \times 10^{-6}$) and distance ($q = 1.6 \times 10^{-3}$) significant, but pointing game not significant after correction ($q = 0.084$). We therefore recommend **layer4 + Grad-CAM++** as the default configuration; it is the best of the six configurations evaluated, without pretense of global optimality.

3.3 Analysis by pathology

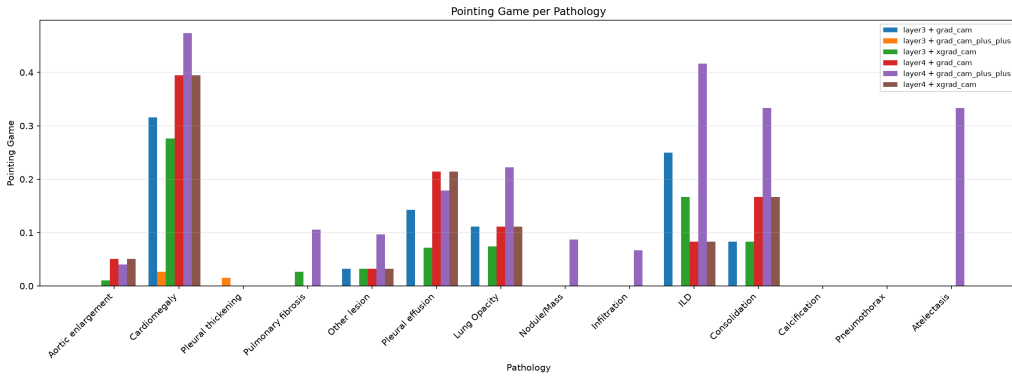


Figure 3: Pointing game by pathology and configuration. Large central pathologies (Cardiomegaly, ILD) are the best localized; focal lesions (Pleural thickening, Calcification, Pneumothorax) are never localized.

Figure 3 and Table 3 break down the pointing game by pathology (layer4 + Grad-CAM++). A clear hierarchy emerges, correlated with the size and position of the lesions: cardiomegaly 0.474 ($n = 76$, CI 0.36–0.58), ILD 0.417 ($n = 12$, CI 0.17–0.67), consolidation 0.333 ($n = 12$). Conversely, pleural thickening ($n = 64$), calcification ($n = 10$) and pneumothorax ($n = 4$) are never localized; nodules/masses 0.087 ($n = 23$). The

small sample sizes of some classes (atelectasis $n = 3$, pneumothorax $n = 4$) make their estimates highly uncertain (wide CIs), unlike Aortic enlargement ($n = 99$) and Pleural thickening ($n = 64$). This gradient suggests that Grad-CAM captures diffuse or central patterns but fails on small peripheral focal lesions.

Table 3: Pointing game by pathology (layer4 + Grad-CAM++). 95% CI by image bootstrap. n = number of findings.

Pathology	n	PG	95 % CI	Pathology	n	PG	95 % CI
Cardiomegaly	76	0.474	0.355–0.579	Other lesion	31	0.097	0.000–0.226
ILD	12	0.417	0.167–0.667	Nodule/Mass	23	0.087	0.000–0.217
Atelectasis	3	0.333	0.000–1.000	Infiltration	15	0.067	0.000–0.200
Consolidation	12	0.333	0.083–0.583	Aortic enlargement	99	0.040	0.010–0.081
Lung Opacity	27	0.222	0.074–0.371	Calcification	10	0.000	0.000–0.000
Pleural effusion	28	0.179	0.036–0.321	Pleural thickening	64	0.000	0.000–0.000
Pulmonary fibrosis	38	0.105	0.026–0.211	Pneumothorax	4	0.000	0.000–0.000

3.4 Controls and randomization test

The table below reports the overall controls (mean score (pointing game + energy in box) / 2, $n = 442$). This combined score is not a conventional metric: it couples the box hitting the argmax (binary) with the proportion of energy inside the box (continuous), taken with equal weight and identically across the four conditions, which makes internal comparisons coherent while preventing a single component from dominating; the two individual metrics are reported elsewhere (Table 2), and the reading caveats below apply to the combined score alone. The trained model (0.080 ± 0.183) is significantly better than the model with randomized weights (0.023, $p = 6.17\text{e-}09$), showing that localization exploits learned features. By contrast, the difference with the untrained model (0.031, $p = 0.36$) and with random maps (0.036, $p = 1.00$) does not reach significance in this test. This result is explained by the very nature of the Mann-Whitney test, which compares entire distributions (stochastic dominance) rather than means: the real scores are extremely asymmetric (median 0.008, 21% zeros), their mean being carried by a small number of successful localizations (cardiomegaly), while random maps produce a quasi-constant floor. The fact that the trained model does not outperform random initialization either does not signify no learning: an untrained convolutional network already generates smooth, centered maps (architectural prior, centered images), whose energy-in-box floor is of the same order; only destroying the learned weights (randomization) lowers this floor significantly. The power of these tests is further limited by the proportion of zeros. Paired tests per finding, better suited to these zero-inflated distributions, could not be computed for lack of saved individual zero scores, and constitute a necessary extension (see the limitations, section 4).

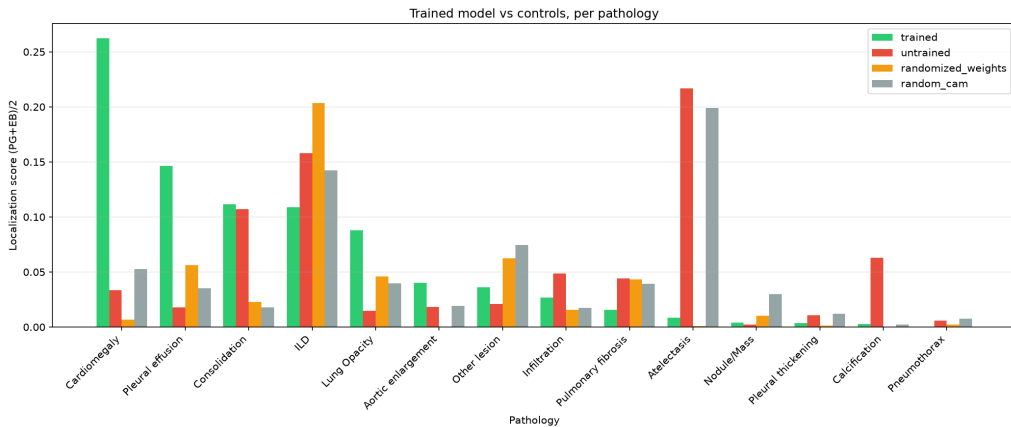


Figure 4: Localization score of the trained model vs three controls, by pathology (Grad-CAM, layer4, combined score). Cardiomegaly is the only class where the trained model clearly exceeds all controls. For ILD, randomized weights exceed the trained model, a sign of an architectural artifact for this configuration (see the caveat of section 3.4).

The per-pathology analysis (Figure 4) substantially nuances the global picture. For **cardiomegaly**, the trained model (0.262) vastly outperforms all controls (randomized: 0.007, $p = 2.86\text{e-}21$), evidencing genuine learned localization. For **ILD**, by contrast, the model with randomized weights (0.204) exceeds the trained model (0.109, $p = 0.96$): for the Grad-CAM and combined score used here, the apparent localization of ILD is an

architectural artifact, not the reflection of learned features. Important caveat: the pointing game of 0.417 in section 3.3 is measured with Grad-CAM++, for which the controls were not recomputed; extending the artifact to Grad-CAM++ is plausible (same architecture, same central prior) but remains to be demonstrated: the conclusion about ILD holds only for the controlled configuration. The 14 tests per condition are corrected with the Benjamini–Hochberg method: after correction, only cardiomegaly (against the three controls, $q \leq 2.7 \times 10^{-8}$), aortic enlargement (against randomized weights and random maps) and pleural thickening (against the untrained model and randomized weights) remain significant; only the cardiomegaly effect survives all conditions. This result, rarely reported, underscores the importance of the weight randomization test before any clinical interpretation of CAM maps.

3.5 Qualitative examples and the No finding class

Figure 5 directly confronts pointing-game successes and failures (Grad-CAM++, layer4). The four successes shown are cardiomegalies: the box, large and centro-mediastinal, contains the map maximum, which concentrates its strongest activation on that region in any case. The four failures are nodules/masses: the box is small and located in the lung periphery, while the map stays centered on the mediastinum and hila; the argmax falls outside the box, sometimes at considerable distance. This gap mechanically illustrates the quantitative analyses of sections 3.3 and 4: the map follows a central spatial prior, which coincides with large central lesions and systematically misses peripheral focal lesions. Figure 6 shows the same pattern on other pathologies: diffuse centro-mediastinal activation regardless of the class, coincidence with the box for cardiomegaly, miss for peripheral lesions (calcification).

For *No finding*, without bounding boxes, the maps are strongly diffuse (spatial entropy ≈ 14.6 – 15.6 bits, close to the maximum), with a concentration in the top 1% of pixels of only 3.6–4.7%. This diffusivity is consistent with the absence of focal lesion, indirectly validating the model’s behavior on this class.

4 Discussion

This study provides a rigorous quantitative validation of Grad-CAM in chest radiography, with three takeaways. First, spatial fidelity depends strongly on the pathology: localization follows a gradient correlated with the size and centrality of the lesion, highest for cardiomegaly (pointing game 0.474, CI 0.36–0.58, therefore still imperfect: the argmax leaves the box in more than half of cases) but nil for nodules and pneumothorax. This gradient echoes the observations of Saporta et al. [2], for whom Grad-CAM failures preferentially concern small, complex-shaped lesions. CAM maps may therefore serve as a clue for diffuse pathologies, but must never be used alone to rule out or localize a micro-lesion. Second, the weight randomization test proves indispensable: without it, ILD localization would have been mistakenly interpreted as a success; the control shows that, at least for Grad-CAM, it stems from an architectural bias, the centro-mediastinal prior, rather than learned features. Extending this control to Grad-CAM++, which carries the pointing game to 0.417, remains to be done and conditions the final scope of this finding (see the limitations). This result calls for caution in a literature where this control is rarely presented, including in recent work that limits itself to qualitative validation of Grad-CAM maps by a clinician [7]. Third, the recommended configuration is layer4 + Grad-CAM++, the best of the six evaluated on all metrics at layer4, the deepest layer carrying the richest semantics.

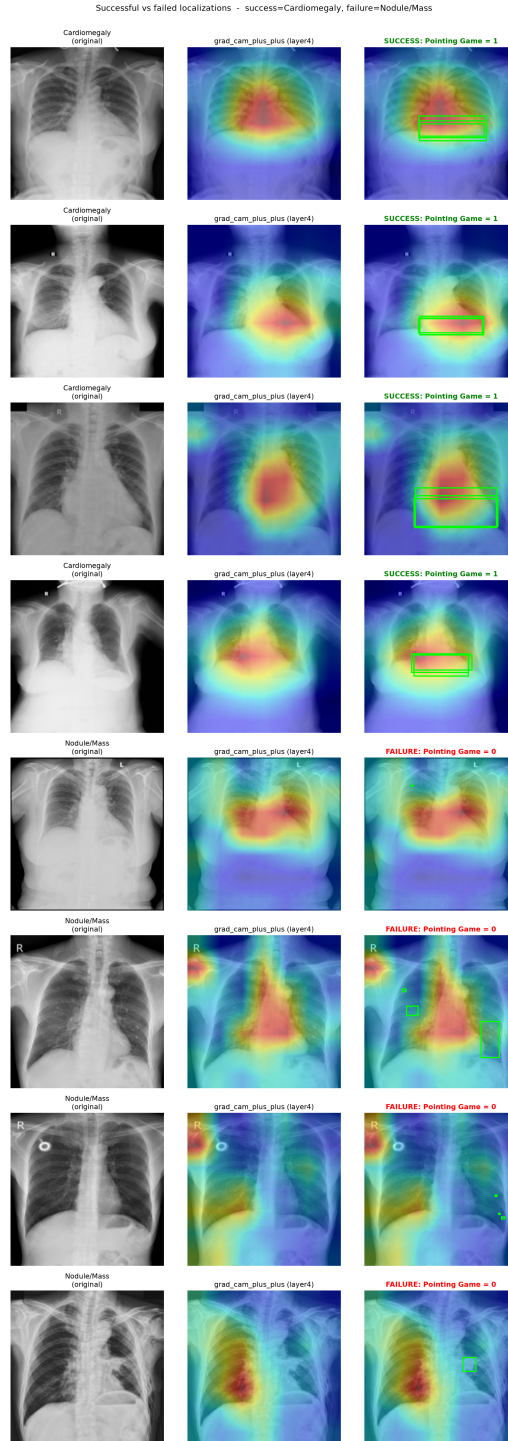


Figure 5: Successes and failures of the pointing game (Grad-CAM++, layer4). Columns: original image, CAM map, map with ground-truth box (green). Rows 1–4: cardiomegalies, successes (PG = 1), the large central box contains the argmax. Rows 5–8: nodules/masses, failures (PG = 0), the map stays central while the box lies in the periphery.

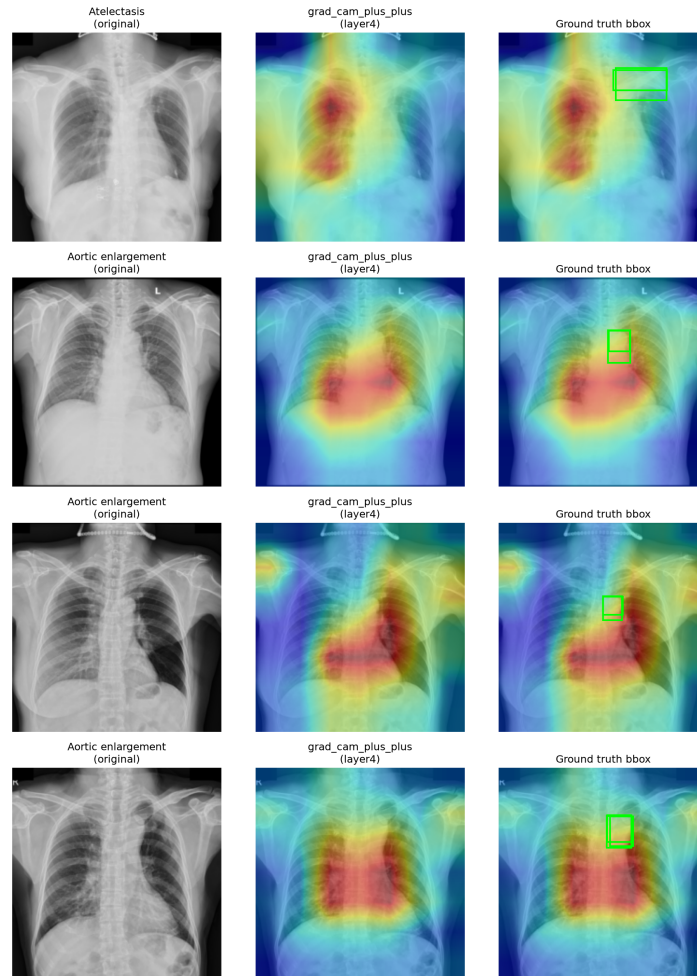


Figure 6: Grad-CAM++ (layer4) examples on other pathologies. Columns: original image, CAM map, map with ground-truth box (green). The diffuse centro-mediastinal activation is systematic and coincides with the box only for large central pathologies.

Limitations

The model is trained from scratch, without ImageNet pretraining, which may underestimate the attainable localization quality. The bounding boxes aggregate several radiologists without explicit modeling of inter-annotator disagreement; retaining the best score among boxes amounts to biasing measurements upward. The metrics assume a rectangular box, a coarse approximation for irregularly shaped lesions; IoU binarized at 0.5 remains arbitrary for uncalibrated maps, and the distance to the center penalizes large boxes even if the maximum is close to it. The global control comparisons rely on an unpaired test (Mann-Whitney); per-pathology tests are corrected with Benjamini–Hochberg, but individual zero scores were not saved, which prevents computing paired tests (Wilcoxon) and per-image bootstrap on the controls, to be produced nonetheless. The controls were computed with Grad-CAM and the combined score, whereas the recommended configuration is Grad-CAM++: the architectural-artifact conclusion for ILD is therefore demonstrated only for Grad-CAM, and its extension to Grad-CAM++ remains to be verified. Classification performance is limited to per-class AUC: average precision, thresholded F1 and calibration, better suited to multi-label imbalance, are not yet measured. Some per-pathology sample sizes are very small (pneumothorax $n = 4$, atelectasis $n = 3$), their CIs being accordingly very wide. A single split (seed=42) and a single training run were used: between-seed variability is not estimated. Three to five retrains would allow quantifying this variance and verifying that the reported gaps (layer4 vs layer3, cardiomegaly vs the other classes) exceed it; otherwise, for lack of compute resources, these gaps must be read as pointwise orders of magnitude, especially for classes with small sample sizes (atelectasis $n = 3$, pneumothorax $n = 4$), whose between-seed variance would by itself be considerable. Finally, the evaluation is limited to a single dataset (VinDr-CXR) and a single architecture (ResNet-50); external validation (CheXpert, NIH ChestX-ray14) would be needed to generalize.

Future work will include using pretrained weights, extending to more recent explainability methods (Score-CAM [8], Layer-CAM [9]), and a clinical evaluation with radiologists measuring the actual impact of these maps on diagnostic decision-making.

5 Conclusion

We quantitatively evaluated the spatial fidelity of Grad-CAM against radiologist annotations on VinDr-CXR. The classifier discriminates well (macro AUC 0.920), but localization is relatively highest for large central lesions (cardiomegaly, pointing game 0.474) while remaining imperfect even for this class, and nearly nil for focal lesions. For ILD, the controls (Grad-CAM) reveal an architectural artifact; extending this to Grad-CAM++, the best-performing configuration, remains to be confirmed. We recommend layer4 + Grad-CAM++ as the default configuration, together with a systematic weight randomization test before any clinical interpretation. These results provide the community with reproducible metrics and concrete recommendations for the responsible use of saliency maps in medical imaging.

6 Code and data

The model source code, evaluation code, statistical analysis scripts (`scripts/`, `src/`) and raw results (`outputs/`) are provided with this report and published on GitHub: <https://github.com/5uru/XLoc-CXR>. VinDr-CXR is a public dataset [3], usable under its original license.

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